CSC340

Introduction

Creating a programing language that is easy to read, write and reliable is what we want in a programing language. So, we must choose the best keyword to describe its functionality, handling the white space and using symbols for operation etc.

Lexer and Parse

1- Lexer

In this project we have written the lexical to make every function like the (for, while, if, else, print) I named everything to its corresponding name and handled the ids it have to start with a letter and then anything else in int the problem that I notice is that if I write it like this (int: [0-9]*) the integer can be empty or start with zero so we write like this (int: [1-9] [0-9]*) this can handle the problem there is another problem I encountered when using (int: [1-9] [0-9]*) and do for loop

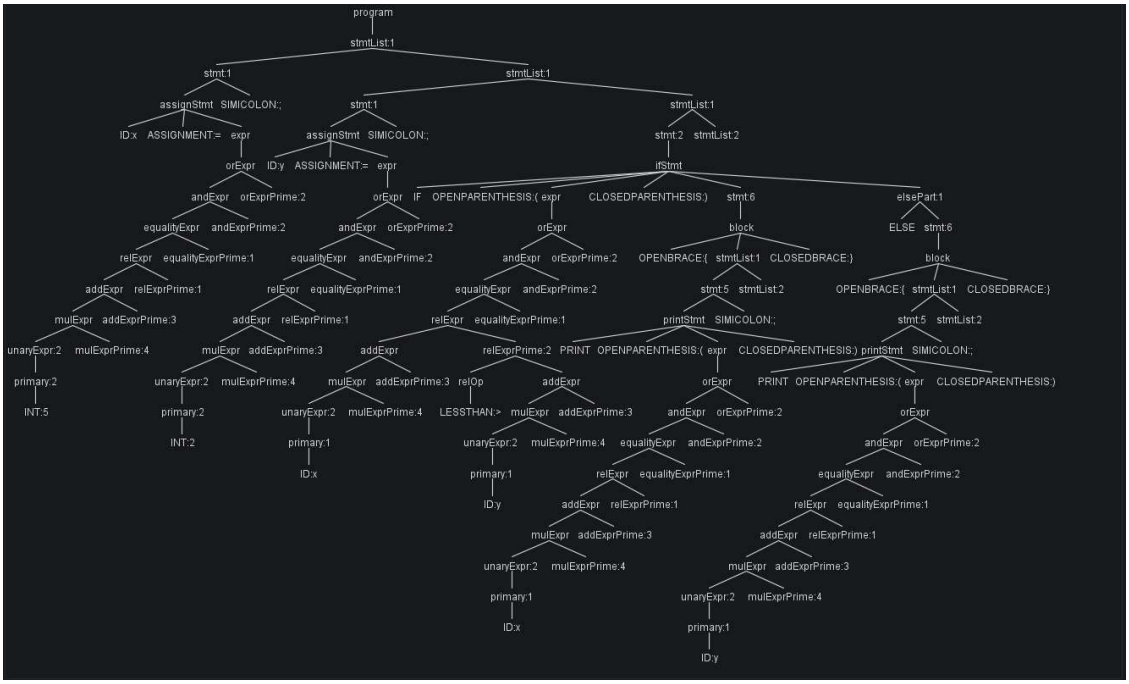
(for (i = 0; i <10; i = i + 1) {print(i);}) i is assigned to zero but the int rule does not allow I to start with zero so we add (INT: '0' | [1-9][0-9]*;) to solve the issue.

2- Parser

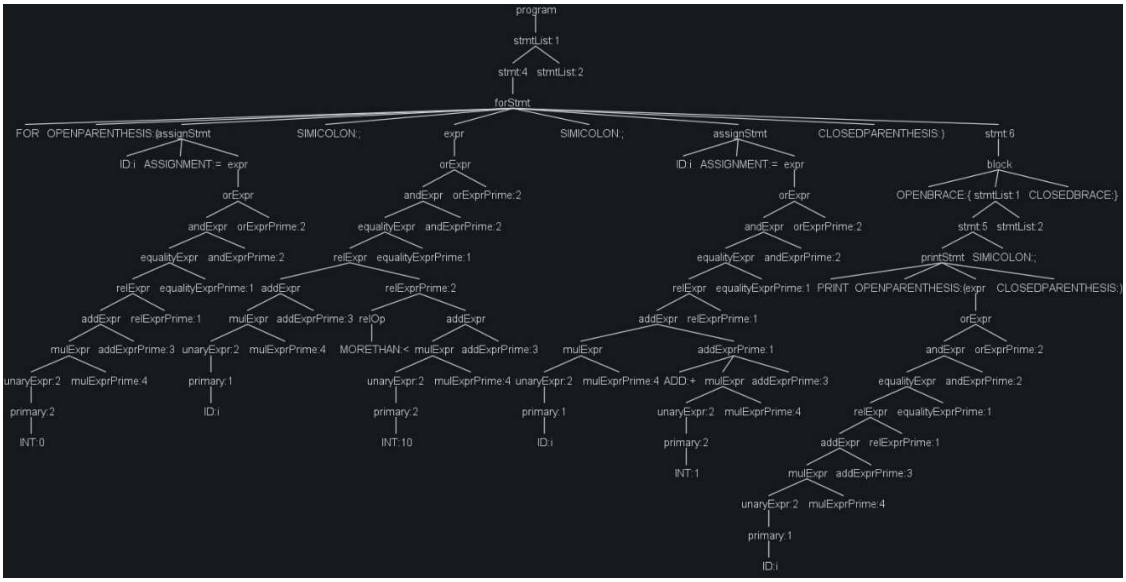
In the parser the only main thing that I noticed was the is left factoring in (EqualityExpr: RelExpr | RelExpr EqOp RelExpr) and (RelExpr : AddExpr | AddExpr RelOp AddExpr) solved it using the left factoring elimination.

Test

```
x = 5;
y = 2;
if (x > y) {
 print(x);
} else {
 print(y);
}
```




```
for (i = 0; i < 10; i = i + 1) {
 print(i);
}
```



Conclusion

In the end every language differs from person to person requirement some prefer readable some prefer writeable some prefer reliable and every language have different usage so use what best for the problem.